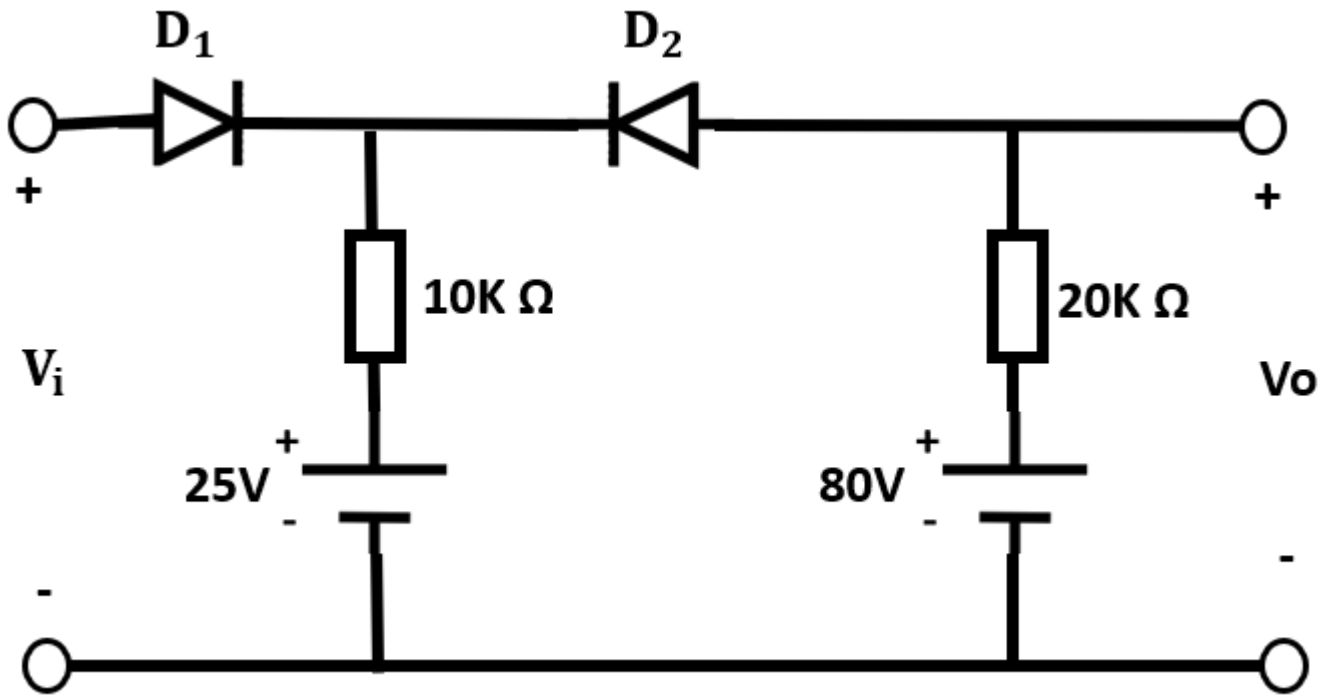
Adrià Barceló Luna
Nil Cosano Alcudia

Exercise 1:

$V_\gamma = 0V$ $R_\gamma = 0\ \Omega$



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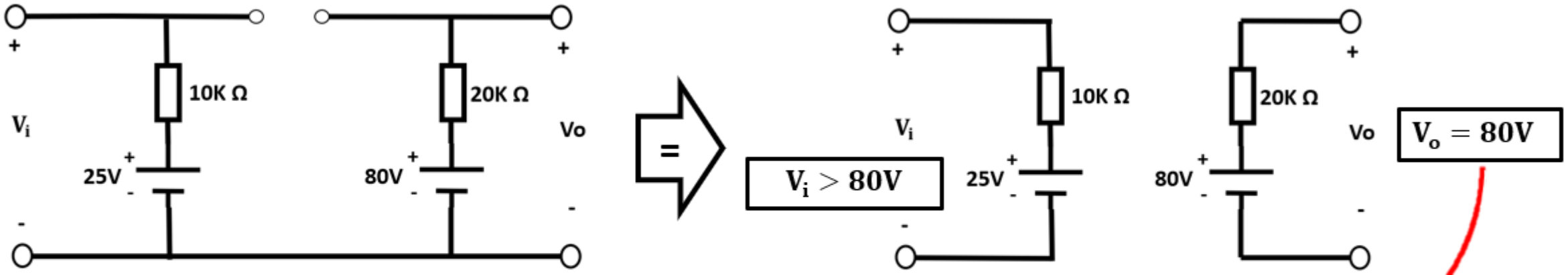
- **Step 1:** $\mathbf{D_1}$ **= ON and** $\mathbf{D_2 = OFF}$ **:**

Si els diodes són ideals:

Change the diodes by if the diode = **ON**
Change the diodes by if the diode = **OFF**



Com V està en paral·lel a la font de 80 V, tindran el mateix valor.

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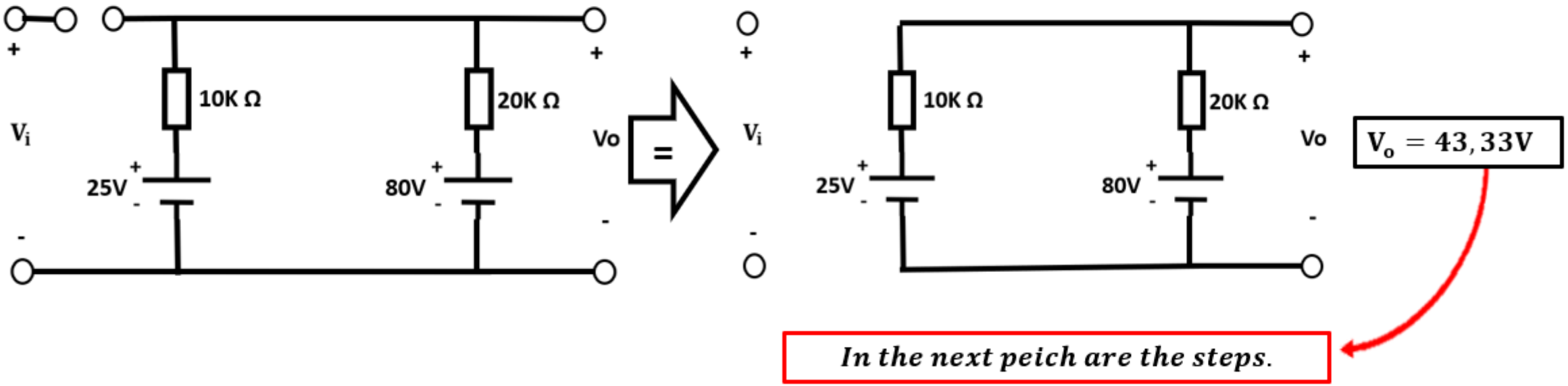
- **Step 2:** $\boxed{D_1 = \text{OFF and } D_2 = \text{ON}}$:

Si els diodes són ideals:

Change the diodes by ○—○ if the diode = **ON**
Change the diodes by ○ ○ if the diode = **OFF**



$V_o = 43,33V$

In the next peich are the steps.

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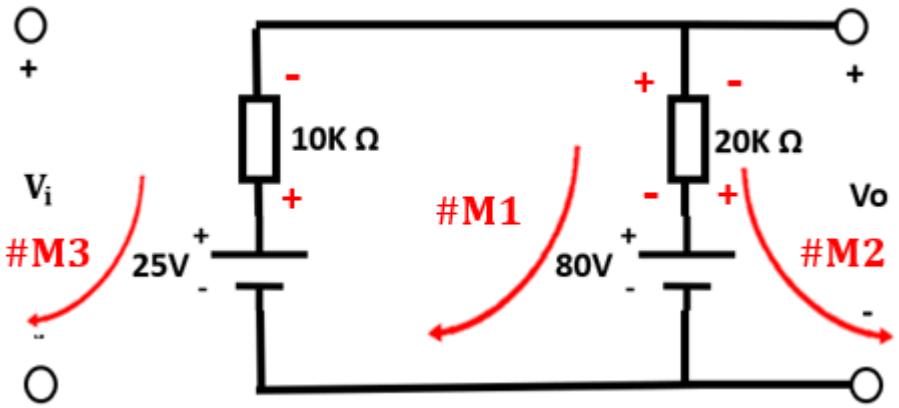
D_1= OFF and D_2 = ON :



Es fa l'anàlisi de les malles :

#Malla 1 (M1):

0 = 80-25+ (10 K+20 K) * I → $I = {}^{-55}/_{30K} = -1,83\ mA$

#Malla 2 (M2):

V_o= 80 - (20k * I) → $V_o = 80 - 36{,}66 = 43,33\ V$

#Malla 3 (M3):

V_i= (10K * I) - 25 → $V_i \leq -43,33\ V$

Nota:

– Si el símbol és negatiu, indica que el corrent circula en sentit contrari a l'escollit al fer l'anàlisi de la malla.

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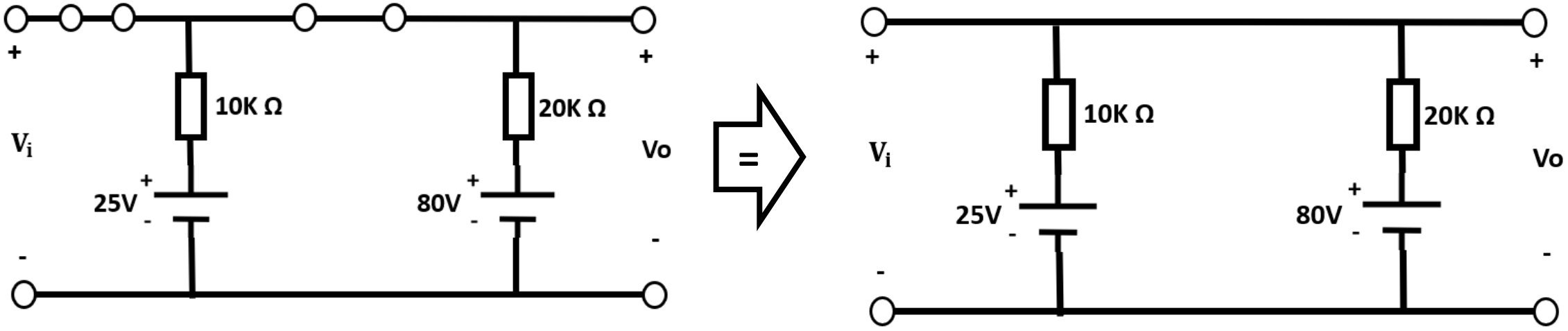
- **Step 3:** $\boxed{D_1 = \text{ON and } D_2 = \text{ON}:}$

Si els diodes són ideals:

Change the diodes by if the diode = **ON**
Change the diodes by if the diode = **OFF**



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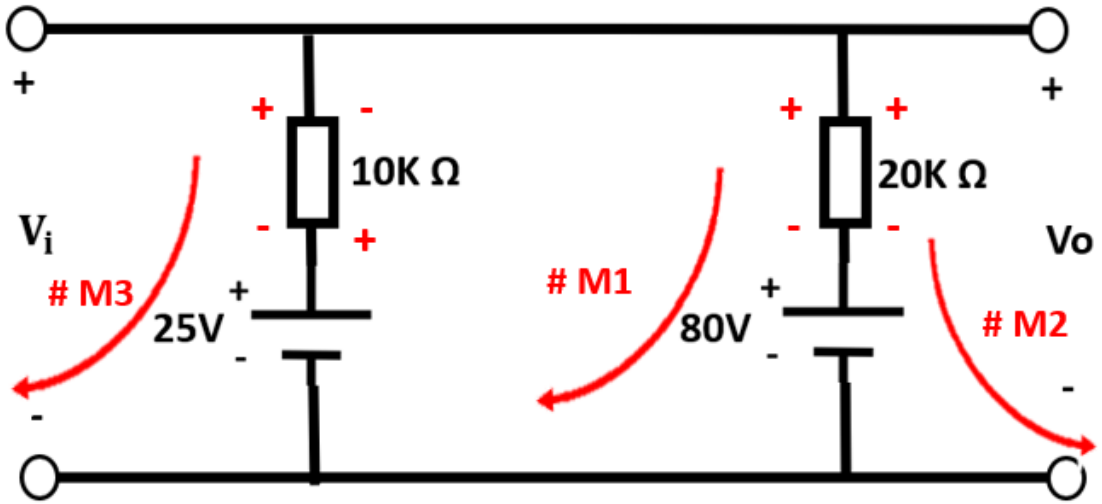
D_1 = ON and D_2 = ON :



Es fa l'anàlisi de les malles :

#Malla 1 (M1):

$$25-80 = [(10\text{ K}+20\text{ K}) * I_2\,] - (\,10\text{K} * I_1) \rightarrow I_2 = \frac{-55 + (\,10\text{K} * I_1)}{30K}$$

#Malla 2 (M2):

$$V_o = 80 - (20\text{k} * I_2\,) = V_i \rightarrow \boxed{V_o = V_i}$$

#Malla 3 (M3):

$$V_i - 25 = (\,10\text{K} * I_1) - (\,10\text{K} * I_2) \rightarrow I_1 = \frac{V_i - 25 + (\,10\text{K} * I_2)}{10K}$$

$$\boxed{\rightarrow 80\text{ V} \geq V_i > 43{,}33\text{ V}}$$